Machine learning in real companies

Real use cases, concepts and trade-offs

InnoIT · 19 October 2026

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Learning path

01. Machine learning foundations

02. Deep learning and image generation

03. GenAI systems and agentic workflows

12 requested topics. 15 cases, a solution diagram for each, and an integrated BBVA Blue closing example.

A guide to the examples

Tasks and implementations

Classification, clustering and forecasting describe tasks. Several algorithm families can solve each one.

RAG and agent orchestration describe systems that combine models with other software.

How to read each case

Each case covers the concept, real workflow, advantages, limitations and a one-sentence takeaway.

Sources describe the documented deployment at the time of publication. Trade-offs are teaching analysis.

01

Machine learning foundations

Relationships, labels, groups and numerical predictions

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Association rules

Walmart (United States)

Production recommendation system

Concept

Find items that appear together more often than chance would suggest.

Real use case

Walmart mines baskets for complementary product pairs and uses the relationships in recommendations.

Advantage

Readable rules can support cross-selling without labeled outcomes.

Limitation

Popularity and promotions distort co-occurrence. A relationship does not prove causation.

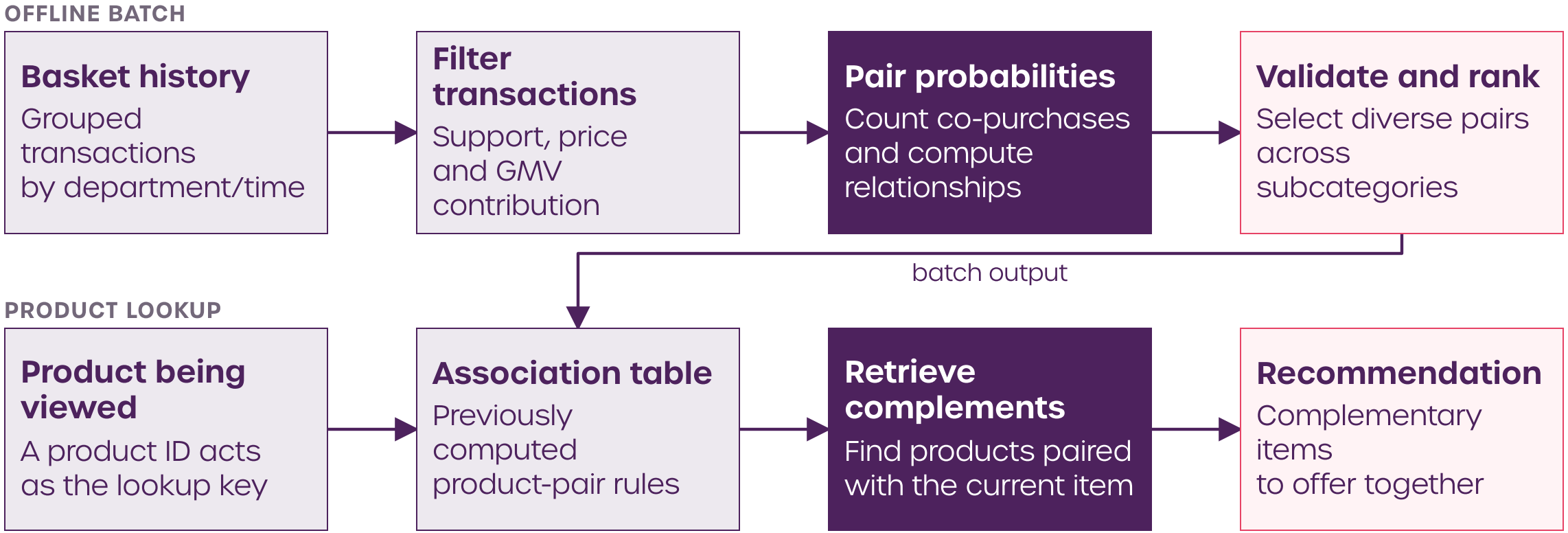
Walmart evaluated FP-Growth, then selected simpler pairwise co-occurrence rules for scale.

Products bought together can inform products recommended together.

Source: Walmart Global Tech

Walmart: basket associations

Batch processing builds the product-pair lookup used by recommendations



The deployed winner uses pairwise probabilities. Walmart evaluated FP-Growth as an alternative.

Source: Walmart Global Tech

Three classification outputs

Task	Labels per observation	Illustrative output
Binary	One of two	Legitimate or bot traffic
Multiclass	One of several	One expense category
Multilabel	Several may apply	Several product interests

"Multicategoría" is treated here as multilabel classification. Examples illustrate the output structure.

Binary classification

Cabify (Spain)

Operational service via Infobip Signals

Concept

Assign one of two labels, often by applying a threshold to a risk score.

Advantage

Automates high-volume decisions and reduces manual intervention.

Real use case

Infobip Signals distinguishes legitimate SMS from bot traffic and blocks fraudulent requests for Cabify.

Limitation

False positives can block real customers. Attack patterns change over time.

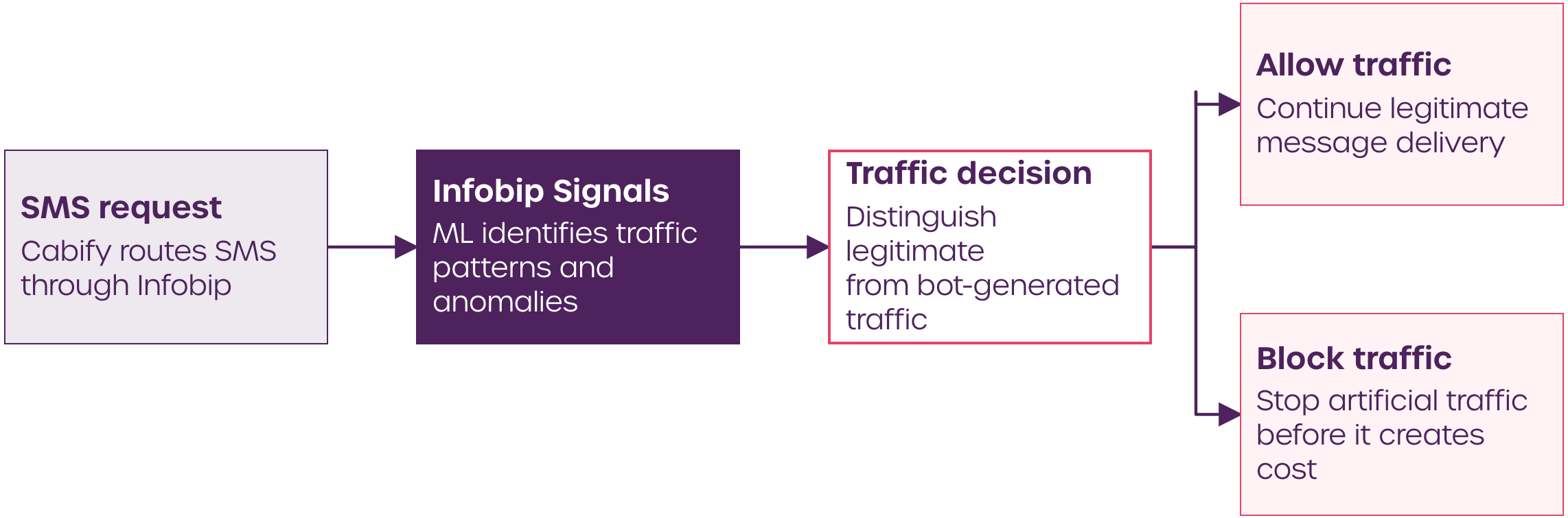
Infobip reports 15.7% less AIT-related financial loss in one quarter. Model architecture is undisclosed.

A two-way prediction becomes useful when its errors have clear business costs.

Source: Infobip customer case with Cabify testimony

Cabify: legitimate or artificial traffic

The ML service makes a concrete allow/block decision on SMS traffic



Infobip discloses the decision and use of ML, but does not publish features, model family or thresholds.

Source: Infobip customer case with Cabify testimony

Multiclass classification

Uber (United States)

COTA support assistant in production (2018 account)

Concept

Choose one category from three or more mutually exclusive alternatives.

Advantage

Reduces the effort of navigating a large support taxonomy.

Real use case

Uber COTA predicts support issue types from ticket text and trip context, helping agents select the right category.

Limitation

Similar issues and rare categories can confuse the classifier.

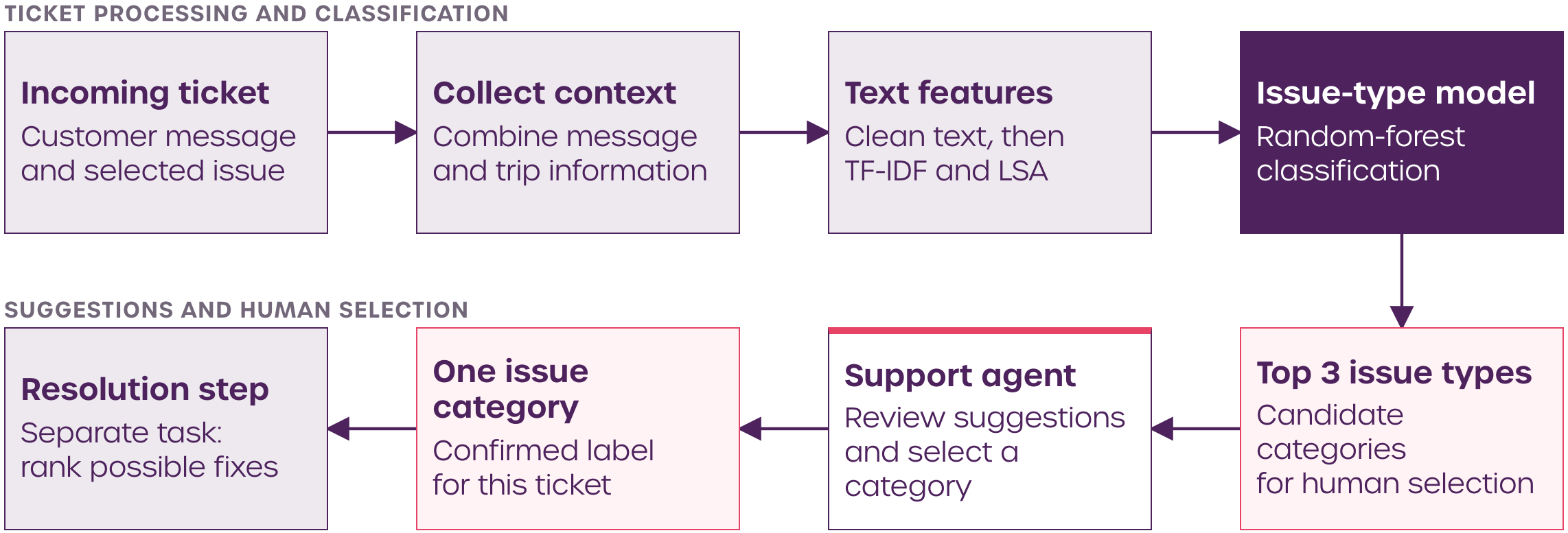
Issue identification is separate from COTA's solution-ranking task.

A ticket gets one issue category, with a human confirming the model's suggestions.

Source: Uber Engineering

Uber COTA: identifying a support issue

Ticket text and trip context help a support agent select one issue category



Issue-type identification is the multiclass task. COTA ranks possible solutions in a separate step.

Source: Uber Engineering

Multilabel classification

Amazon Retail (United States)

Production customer propensity inference

Concept

Assign zero, one or several labels to the same observation.

Advantage

Represents overlapping interests without forcing a single segment.

Real use case

Amazon predicts customer action propensities across thousands of product categories, allowing overlapping interests.

Limitation

Rare labels and different thresholds make training and evaluation harder.

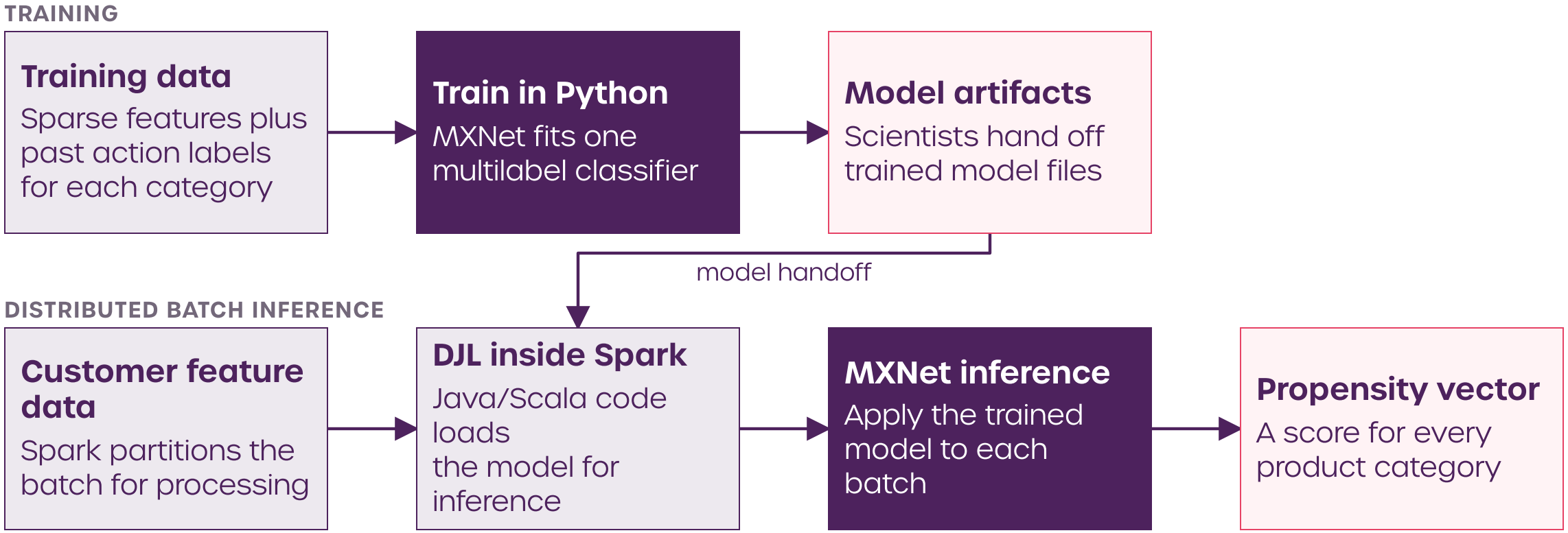
Amazon describes MXNet models with Spark and Deep Java Library inference.

The same customer can belong to several useful categories at once.

Source: Amazon / AWS Open Source Blog

Amazon: thousands of propensities

One sparse customer vector feeds a shared model with many category outputs



Several categories can receive high propensity at the same time. The system does not select one winning label.

Source: Amazon / AWS Open Source Blog

Clustering

Spotify (Sweden)

Documented user-research method

Concept

Discover groups with similar behavior without predefined group labels.

Advantage

Finds unexpected segments and supports targeted user research.

Real use case

Spotify combines UMAP with recursive HDBSCAN to discover finer segments of users.

Limitation

Feature choices shape the groups. Stable clusters still need human validation.

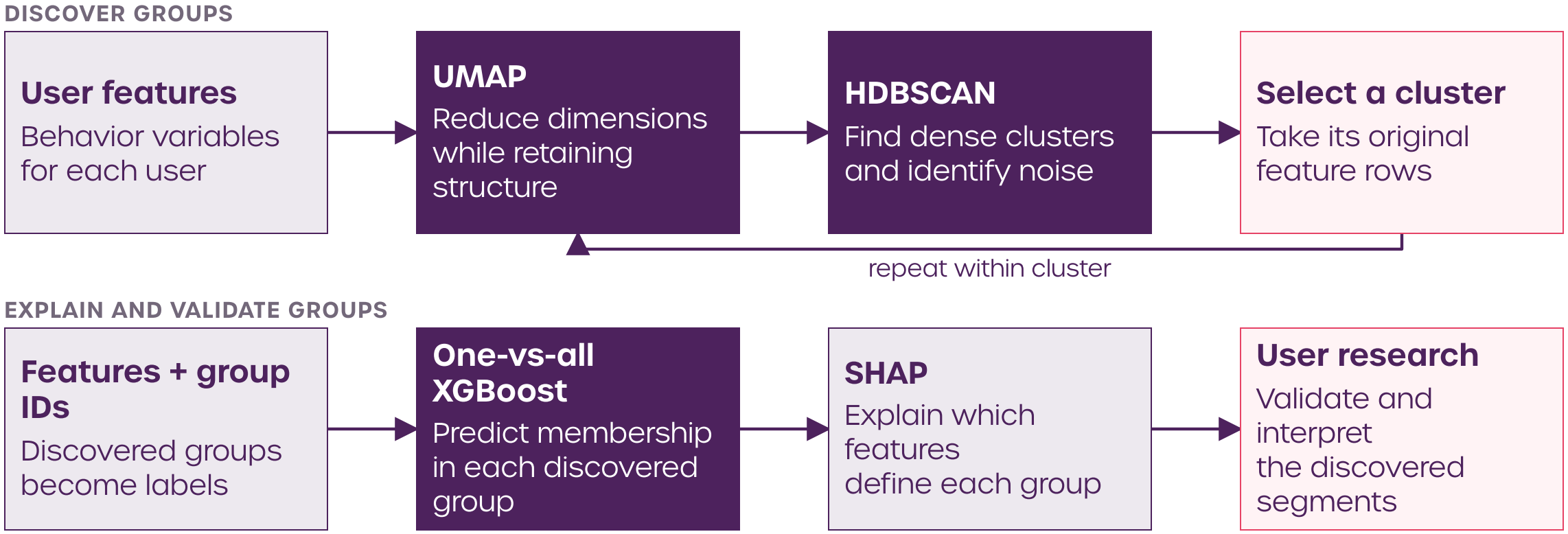
Spotify uses XGBoost and SHAP to explain the resulting segments.

Clustering discovers groups that researchers can then investigate.

Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster



Cluster discovery and supervised explanation serve different purposes. The source describes user research.

Source: Spotify Engineering

Anomaly detection

Adyen (Netherlands)

Production payment-volume monitoring

Concept

Flag observations that differ from the expected pattern for their context.

Advantage

Can identify rare operational problems without labeled incident examples.

Real use case

Adyen models merchant payment volumes and alerts teams when observations fall outside expected bounds.

Limitation

Holidays and merchant changes cause false alerts. Sensitive thresholds increase noise.

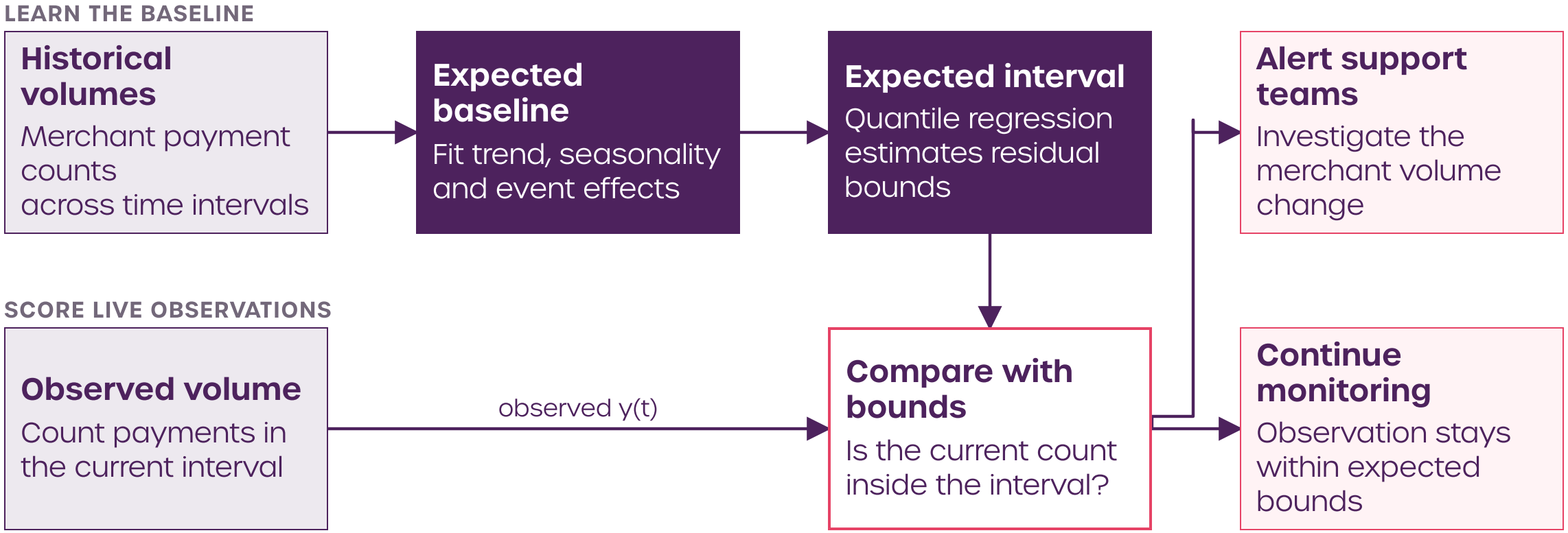
The model captures trend, seasonality and events, with quantile regression for uncertainty bounds.

An anomaly is unusual relative to a meaningful baseline.

Source: Adyen Engineering

Adyen: detecting a payment-volume drop

A contextual baseline and uncertainty bounds define what counts as unusual



A low absolute count can be normal. The decision depends on the merchant, time and expected interval.

Source: Adyen Engineering

Regression analysis

idealista/data (Spain)

Commercial automated valuation service

Concept

Estimate a continuous number from explanatory features.

Real use case

idealista/data combines comparables and hedonic ML to estimate property values from location and attributes.

Advantage

Produces consistent estimates across large property portfolios.

Limitation

Missing attributes and market changes reduce accuracy. Estimates differ from completed sale prices.

The company discloses the valuation approach, but not the exact estimator.

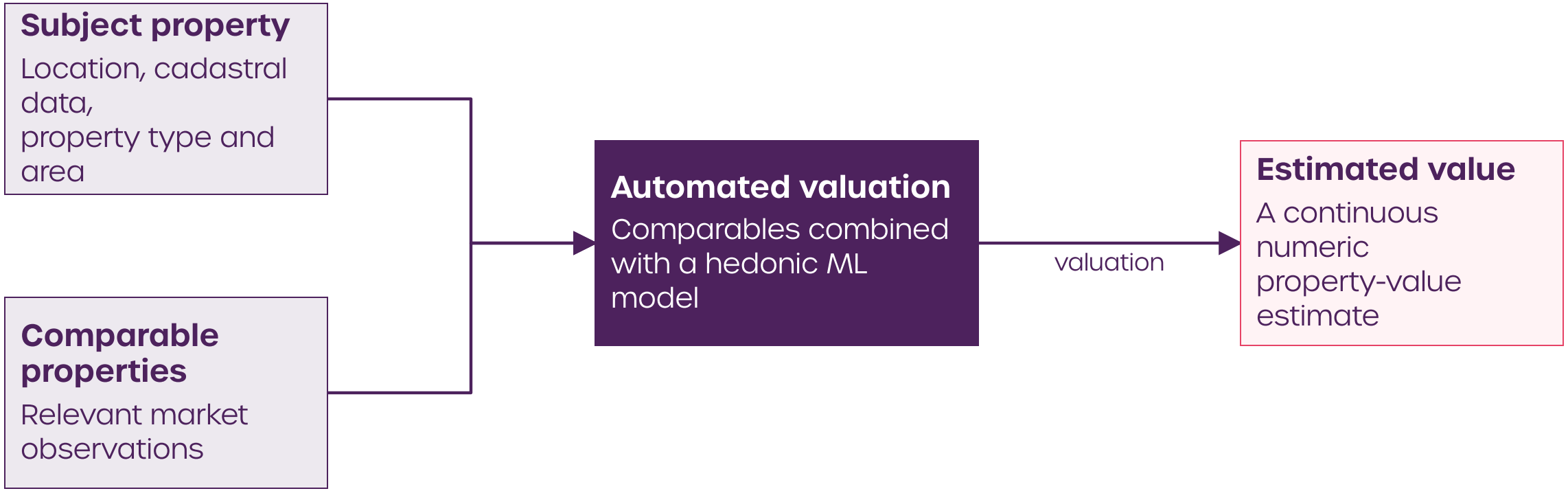
Regression estimates a number from the characteristics of an observation.

Source: idealista/data and idealista labs

idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data



The published approach combines comparables and hedonic ML. The exact estimator and weighting are not public.

Source: idealista/data and idealista labs

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Forecasting

Iberdrola (Spain)

Operational MeteoFlow system

Concept

Estimate future values using time-dependent patterns and available drivers.

Real use case

MeteoFlow combines weather and generation data, using ML to refine renewable output forecasts.

Advantage

Supports planning when wind and solar output vary with weather.

Limitation

Forecast uncertainty grows with the horizon and unexpected weather changes.

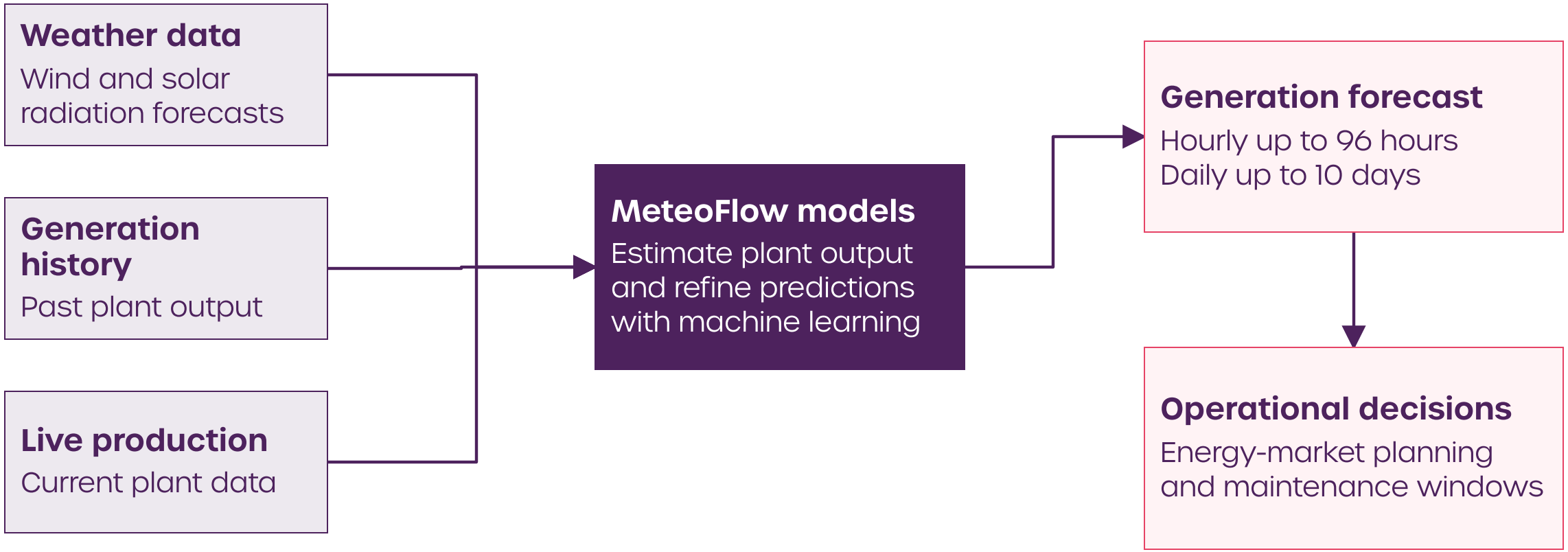
Iberdrola documents 450+ wind and solar plants, with hourly forecasts up to 96 hours.

Forecasting turns an uncertain future into a quantified planning input.

Source: Iberdrola

Iberdrola: renewable output forecasts

MeteoFlow combines external weather with historical and live plant information



The data sources and forecast horizons are documented. Iberdrola does not name the exact ML model family.

Source: Iberdrola

Sentiment analysis and NLP

Booking.com (Netherlands)

Documented review-analysis workflow

Concept

Extract opinions from language and link each opinion to what it describes.

Advantage

Makes specific praise and complaints visible across many reviews.

Real use case

Booking.com links review opinions to 50+ aspects, such as cleanliness or location.

Limitation

Sarcasm, mixed opinions and language differences can change interpretation.

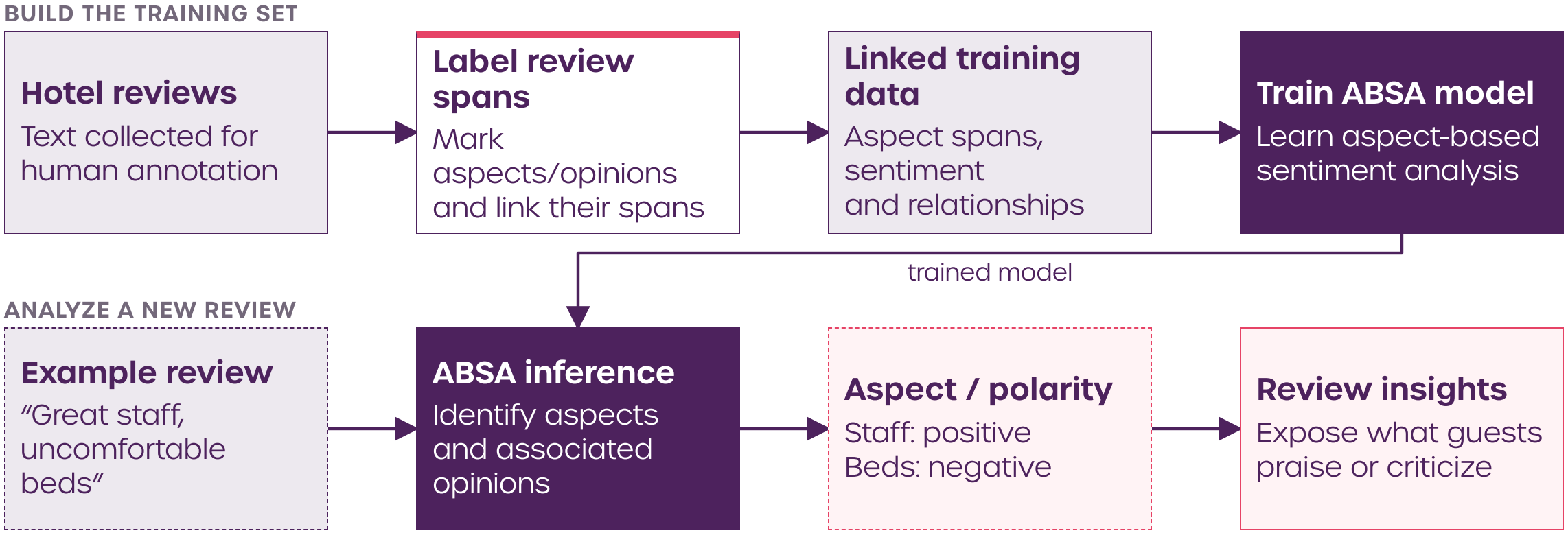
Aspect extraction identifies the subject. Sentiment analysis estimates the opinion about it.

A review can praise the location while criticizing the room.

Source: Booking.com team / AWS Machine Learning Blog

Booking.com: aspect-level opinions

Training links aspect spans to sentiment. Inference returns separate judgments



Dashed boxes contain a teaching example. The article documents annotation and model purpose, not the model architecture.

Source: Booking.com team / AWS Machine Learning Blog

Text mining

ING (Netherlands)

Deployed credit-risk monitoring application

Concept

Extract structured facts and recurring signals from a large text collection.

Real use case

ING's Early Warning System combines signals from news and market data for credit-risk analysts.

Advantage

Helps analysts monitor more external information than they can read manually.

Limitation

Irrelevant or duplicated news creates noise. A signal still needs an analyst's judgment.

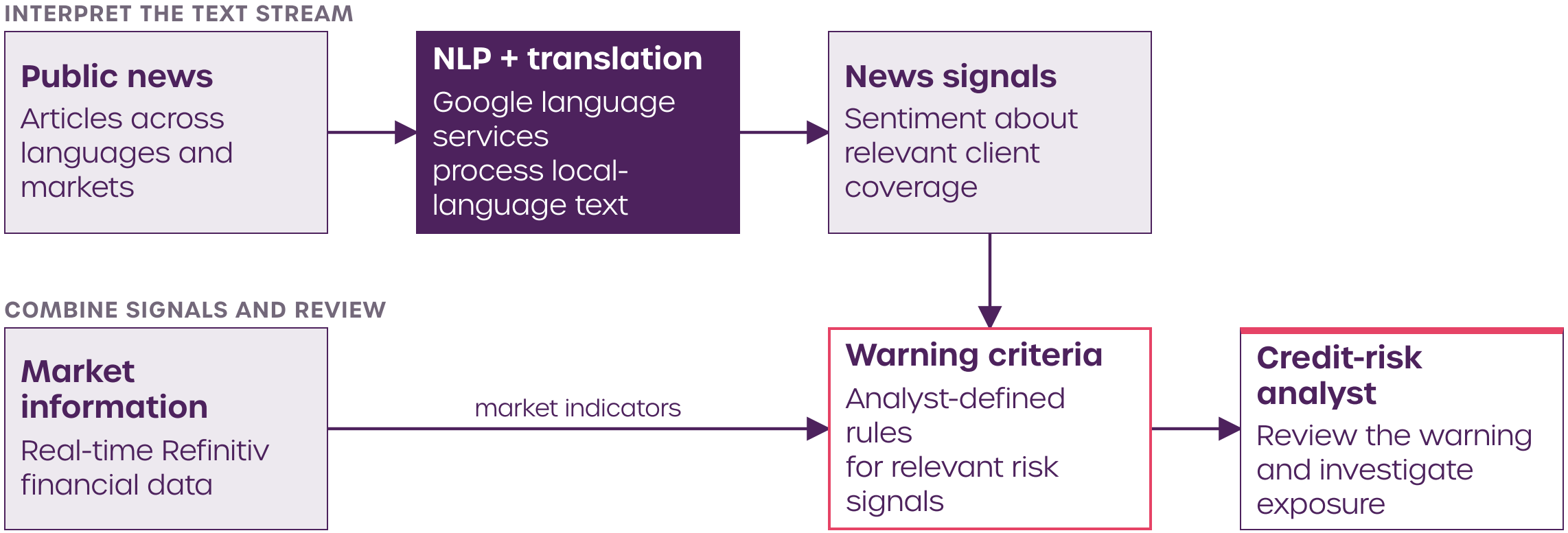
ING describes processing up to 80,000 news articles per day.

Text mining turns a document stream into evidence for investigation.

Source: ING

ING: news becomes a risk watchlist

News analysis and market signals meet analyst-defined warning criteria



The output is an analyst warning. The source presents default prediction as a future ambition.

Source: ING

02

Deep learning

Neural networks learn visual features and generate new images

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Pattern recognition

BMW Group (Germany)

Series-production applications since 2018

Concept

Learn visual features that identify an object or a recurring configuration.

Advantage

Performs repeated visual checks at production speed.

Real use case

At Dingolfing, BMW compares a vehicle's model badge with its order specification and flags missing or incorrect badges.

Employees label component images and train a neural network.

Limitation

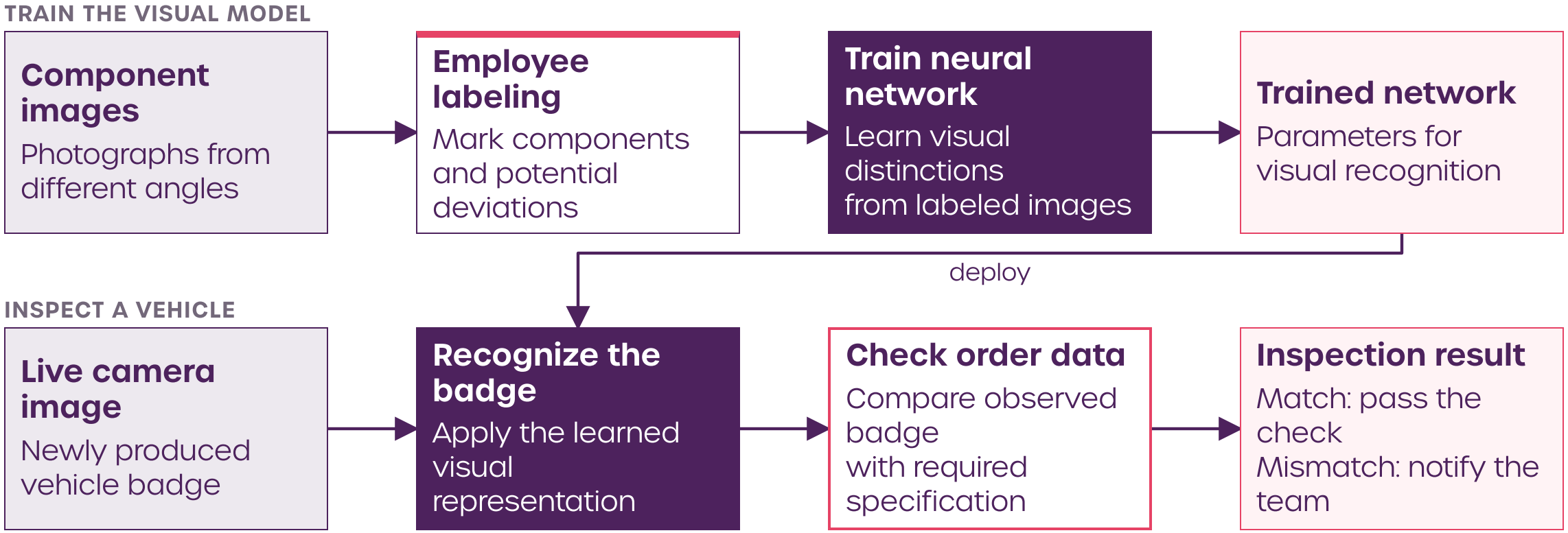
New components and lighting conditions require representative images and monitoring.

A learned visual pattern can turn a camera image into a quality check.

Source: BMW Group

BMW: image recognition checks the badge

Learning the visual pattern and checking the order are separate steps



The neural network recognizes the image. Business logic compares the recognition result with vehicle order data.

Source: BMW Group

BMW visual inspection

The camera observes a vehicle component.

The neural network recognizes it

Software compares the result with the order specification and flags a mismatch.



BMW Group production photo. Source: BMW Group, 15 July 2019.

Image generation

Mango (Spain)

Published Mango Teen campaign

Concept

Generate new images using learned visual structure.

Advantage

Expands creative options and supports repeatable production workflows.

Real use case

Mango created Sunset Dream using real garment photos, a generative model and human selection and retouching.

Limitation

Product fidelity and editorial quality still require human art direction.

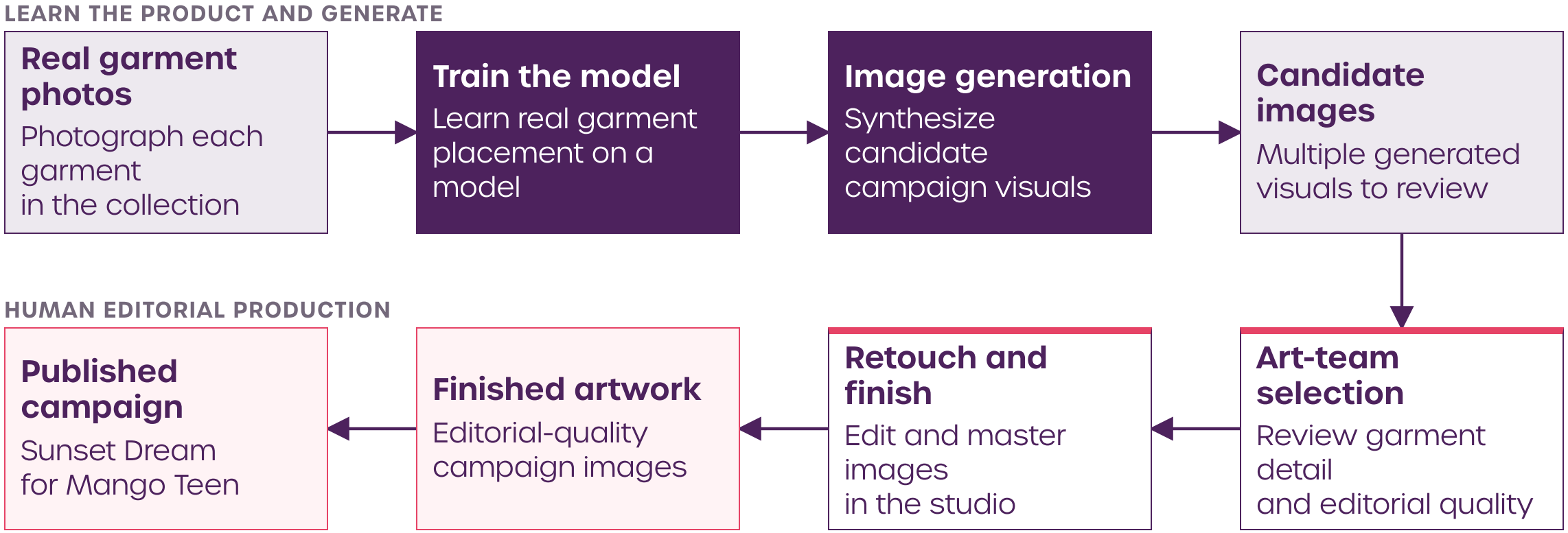
Mango names no specific model architecture. Diffusion and GANs are general examples, not confirmed choices.

Generate new imagery while preserving the real product's characteristics.

Source: Mango Fashion Group

Mango: real garments to campaign images

The published workflow includes model training and human editorial production



Mango does not name diffusion, GANs or a specific model. Human selection and retouching are documented steps.

Source: Mango Fashion Group

Mango Teen: Sunset Dream

A published campaign created with generative AI in July 2024.

Real garment photographs anchor the work

Mango's art team selects and retouches generated images to preserve product detail and editorial quality.



Official campaign image. Credit: Mango Fashion Group.

03

GenAI systems

Retrieval, multimodal generation and coordinated tool use

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What RAG adds to a language model

Retrieval

1. Read the employee's question.
2. Search the authorized knowledge base.
3. Select relevant document passages.

Grounded generation

4. Give the model the retrieved evidence.
5. Generate an answer with sources.
6. Check the answer or escalate.

Intelligent search with RAG

Allianz (Germany)

EKA live at Allianz Leben from July 2024

Concept

Retrieve relevant knowledge before an LLM generates an answer.

Advantage

Makes internal knowledge easier to access and inspect.

Real use case

Allianz EKA searches AWISO, summarizes relevant knowledge and links call-center staff to full sources.

Limitation

Missing retrieval results and outdated documents can still produce wrong answers.

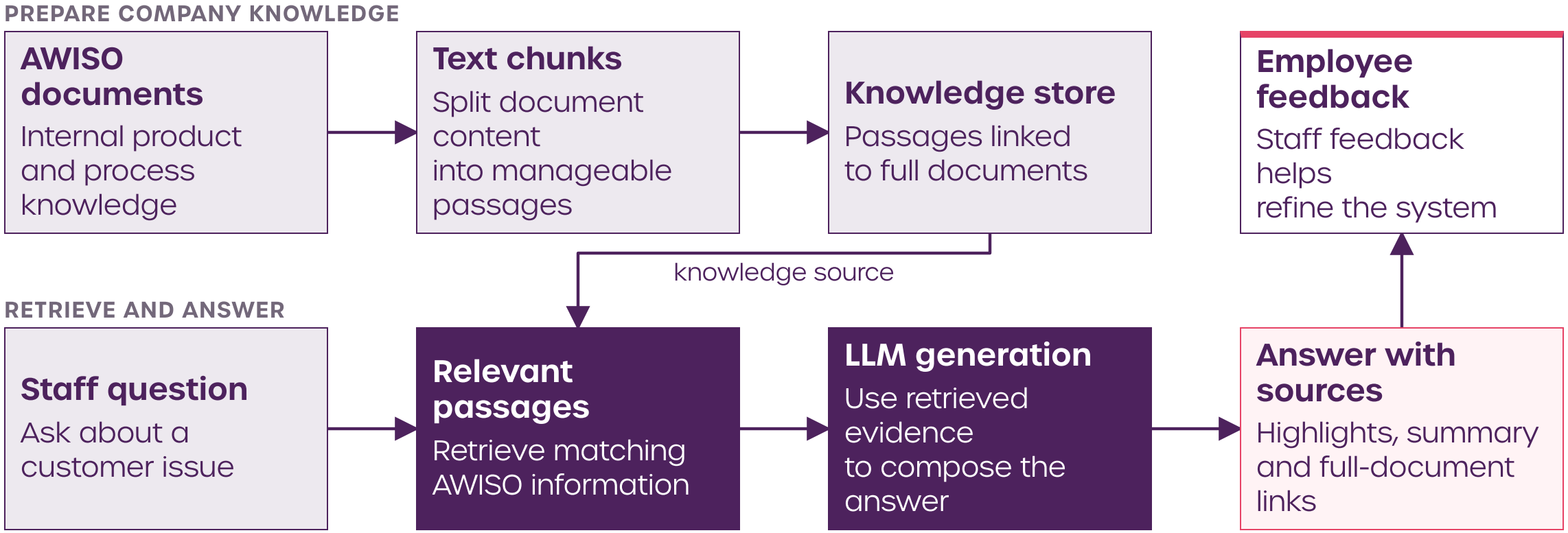
Allianz reports over 90% accuracy on common queries, without publishing the benchmark methodology.

Retrieve the evidence first, then generate a grounded answer.

Source: Allianz

Allianz: retrieval grounds the answer

EKA searches AWISO knowledge and returns inspectable information to staff



This functional RAG view omits EKA's other agents. The source does not publish its exact retriever or embedding model.

Source: Allianz

Multimodal generation

Google (United States)

Veo 3 launched in Google Vids (June 2025)

Concept

Generate related content across modalities, such as video and sound.

Real use case

Google Vids uses Veo 3 to turn a text prompt into a video clip with sound, ready to preview and insert.

Advantage

Creates an audiovisual draft without filming or recording a first version.

Limitation

Visual continuity, speech and factual details still need human review.

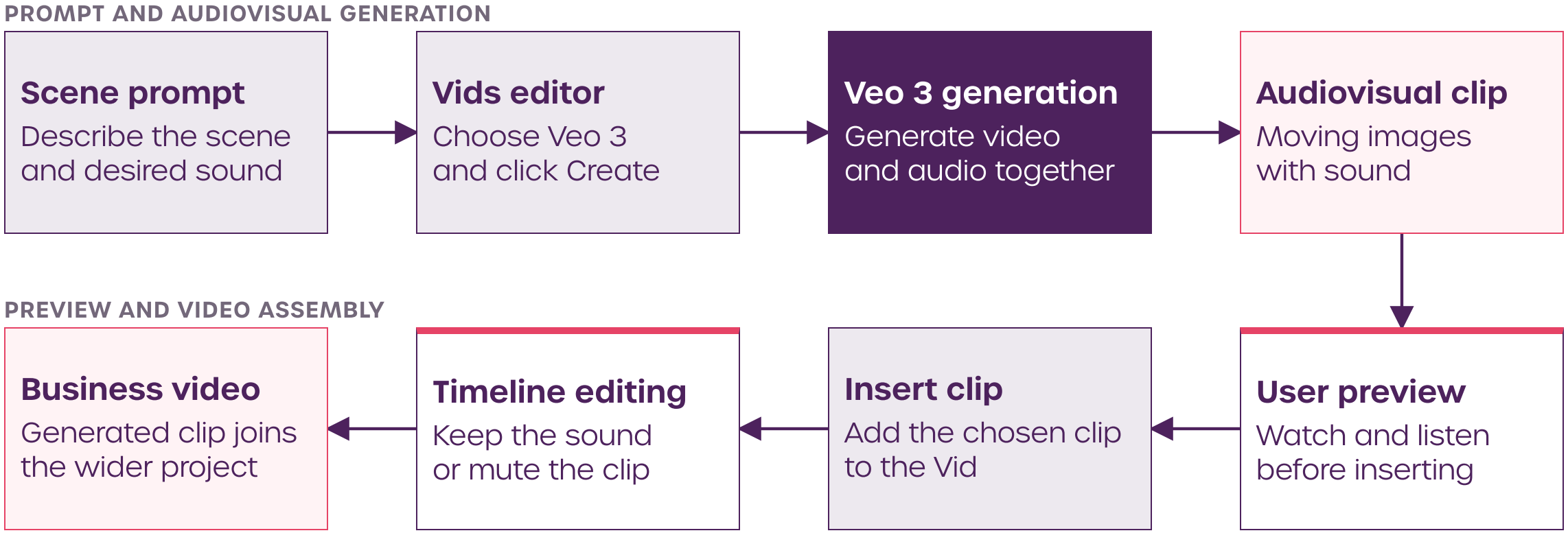
The launch release documents 8-second clips. This is a shipped product capability.

One prompt can generate both the moving scene and its accompanying sound.

Source: Google Workspace Updates

Google Vids: generating video and sound

Veo 3 generates both modalities. The user previews the clip before inserting it



The June 2025 release documents 8-second clips with sound. Preview and insertion remain user-controlled.

Source: Google Workspace Updates

What an orchestrator coordinates

Request and routing

1. Interpret the goal and context.
2. Select a relevant specialist or tool.
3. Request clarification when needed.

Action and feedback

4. Execute the authorized operation.
5. Inspect the result and update state.
6. Reply, retry or hand over to a person.

Agentic workflow orchestration

BBVA (Spain)

Blue assistant deployed in Spain in 2025

Concept

Coordinate specialized agents and tools to complete a bounded task.

Advantage

Connects conversational requests to defined account and card operations.

Real use case

Blue routes requests to a banking tool, an information agent or a clarification step.

Limitation

Routing errors can select the wrong workflow. Permissions and confirmations matter.

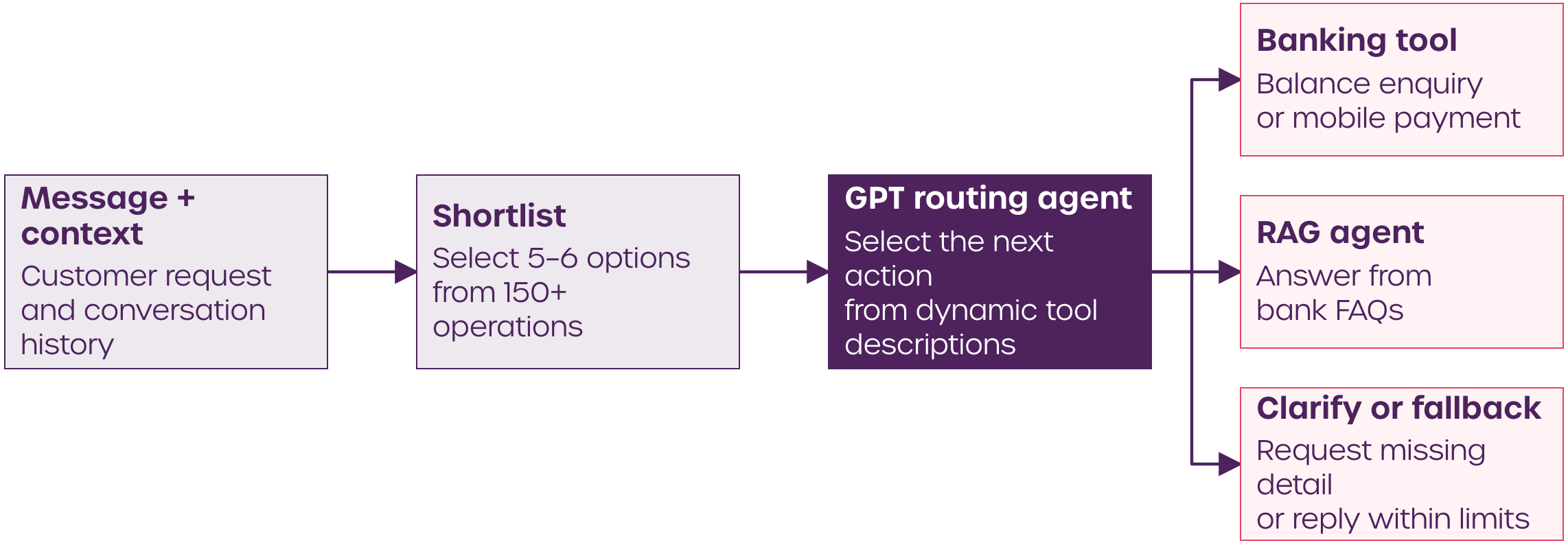
BBVA describes filtering 150+ operations to around 5–6 candidates before routing.

Agents become operationally useful when the right tool can carry out the task.

Source: BBVA AI Factory and BBVA rollout announcement

BBVA Blue: routing a banking request

The routing agent selects a specialist, a tool or a clarification response



Banking operations are callable tools. A conversation can branch between information, action and clarification.

Source: BBVA AI Factory and BBVA rollout announcement

Method selection: structured outputs

Business question	Method	Case
What occurs together?	Association rules	Walmart
Which of two outcomes?	Binary classification	Cabify
Which single category?	Multiclass classification	Uber
Which overlapping labels?	Multilabel classification	Amazon
Which groups emerge?	Clustering	Spotify
What deviates from normal?	Anomaly detection	Adyen
What numeric value?	Regression	idealista
What future value?	Forecasting	Iberdrola

The output and business decision determine the task.

Method selection: language and GenAI

Business question	Method	Case
What is the opinion about?	Sentiment and NLP	Booking.com
What signals occur in text?	Text mining	ING
What visual pattern is present?	Pattern recognition	BMW
What image can we create?	Image generation	Mango
What do our documents say?	Search with RAG	Allianz
Can a prompt yield video and sound?	Multimodal generation	Google
Can software complete the task?	Agent orchestration	BBVA

Image generation uses deep learning and is already a form of GenAI.

Business value and model quality

A useful model changes a decision or makes a workflow easier.

- **Classification:** precision and recall reflect different error costs.
- **Prediction:** backtests need realistic time and data boundaries.
- **GenAI systems:** correct retrieval or fluent text alone does not prove task success.
- **Every method:** deployment needs ownership, monitoring and a clear fallback.

Sources 1 of 3

Walmart Association rules

Walmart Global Tech · 24 Aug 2022

Cabify Binary classification

Infobip customer case with Cabify testimony · Undated customer case

Uber Multiclass classification

Uber Engineering · 3 Jan 2018

Amazon Retail Multilabel classification

Amazon / AWS Open Source Blog · 14 Jul 2020

Spotify Clustering

Spotify Engineering · 5 Dec 2023

Primary company or implementation-partner sources. Additional links appear in speaker notes. Checked 19–20 October 2026.

Sources 2 of 3

Adyen Anomaly detection

Adyen Engineering · 9 Oct 2019

idealista/data Regression analysis

idealista/data and idealista labs · Product page and 17 Feb 2021 engineering post

Iberdrola Forecasting

Iberdrola · Undated corporate project page

Booking.com Sentiment analysis and NLP

Booking.com team / AWS Machine Learning Blog · 14 Jan 2022

ING Text mining

ING · 17 Dec 2018

Primary company or implementation-partner sources. Additional links appear in speaker notes. Checked 19–20 October 2026.

Sources 3 of 3

BMW Group Pattern recognition

BMW Group · 15 Jul 2019

Mango Image generation

Mango Fashion Group · 10 Jul 2024

Allianz Intelligent search with RAG

Allianz · 7 Feb 2025

Google Multimodal generation

Google Workspace Updates · 9 Jun 2025

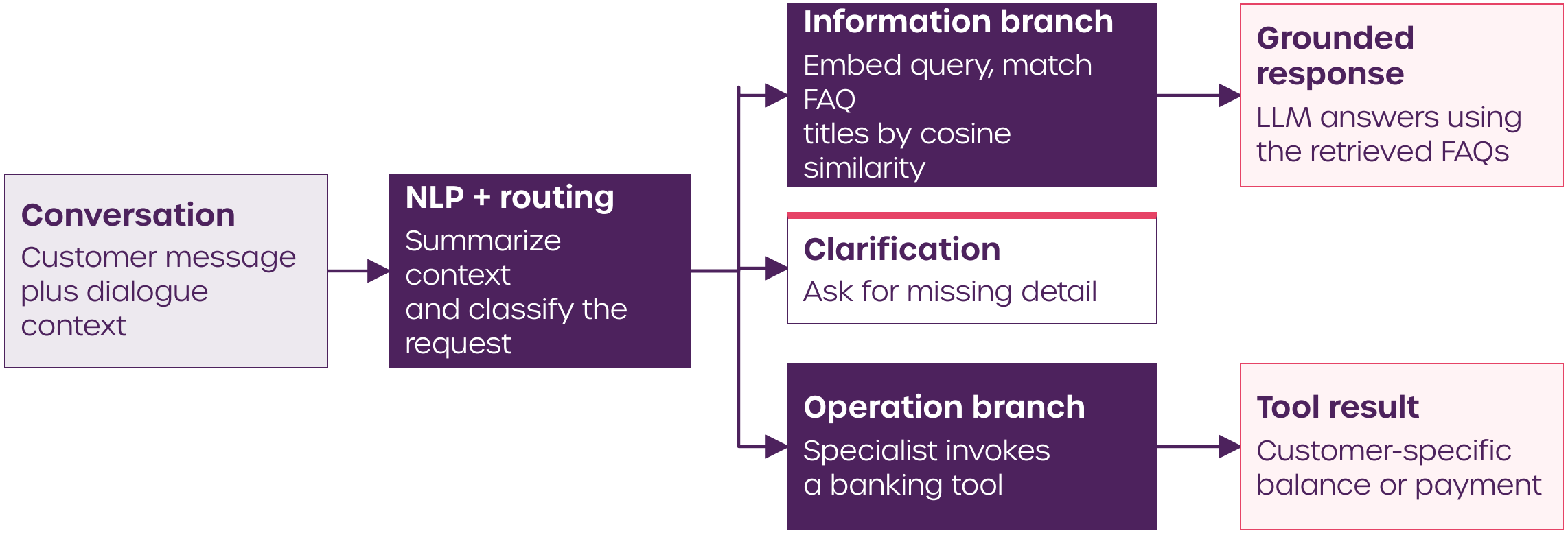
BBVA Agentic workflow orchestration

BBVA AI Factory and BBVA rollout announcement · 13 Nov 2025 technical account

Primary company or implementation-partner sources. Additional links appear in speaker notes. Checked 19–20 October 2026.

BBVA Blue: knowledge and actions

One assistant combines NLP, intent classification, RAG and tool orchestration



Generic FAQs and customer-specific operations follow different branches. Missing information triggers clarification.

Source: BBVA AI Factory: routing and RAG information agent

Discussion

Which business decision needs a better input?

A rule, a prediction, generated content or a completed task?

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